Murtadha Alsayegh

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Summary of Qualifications

Dynamic and solution-oriented Ph.D. in Computer Science with expertise in robotics, artificial intelligence, and privacy preservation within AI systems. Proven track record in advancing secure robotic communications and operations. Experienced educator with a strong background in teaching AI and robotics concepts to undergraduate and high school students. Skilled in leading cross-functional teams to develop innovative solutions that enhance data privacy and operational efficiency. Committed to driving cutting-edge research and contributing to academic and industry advancements through collaborative and impactful projects.

Education

Ph.D. in Computer Science July 2024
Florida International University, Miami, FL
M.S. in Software Engineering April 2013
University of Michigan, Dearborn, MI
B.S. in Computer Science December 2010
Lawrence Technological University, Southfield, MI

Research Interests

Artificial Intelligence, Multi-Agent Systems, Robotics, Reinforcement Learning, Distributed Autonomous Systems, Privacy-Preserving AI, Secure Multi-Party Computation, Motion Planning, Human-Robot Collaboration, Agricultural Robotics.

Professional Experience

Graduate Faculty Scholar Jan 2026 – Present
University of Central Florida, Orlando, FL

- Appointed as a Graduate Faculty Scholar in recognition of contributions to Modeling and Simulation research and graduate education.
- Mentor graduate students conducting research in robotics, artificial intelligence, multi-agent systems, and secure autonomous systems.
- Collaborate with faculty across disciplines on research initiatives and graduate-level academic activities.
- Contribute to the development of graduate research programs through student supervision, research guidance, and scholarly collaboration.
- Support UCF's partnership-driven research mission through interdisciplinary projects and external collaborations.

Postdoctoral Scholar Sep 2024 – Dec 2025
University of Central Florida, Orlando, FL

- Led research integrating artificial intelligence, secure computation, and distributed control for autonomous and cyber-physical systems.
- Developed privacy-preserving algorithms and secure communication protocols for multi-agent robotic systems.
- Mentored graduate and undergraduate researchers, supporting their technical and professional development.
- Collaborated with interdisciplinary teams on USDA-funded agricultural robotics and intelligent automation projects.
- Contributed to proposal development and grant writing activities supporting federally funded research initiatives.
- Published research findings in leading IEEE conferences and journals in robotics, artificial intelligence, and autonomous systems.

Graduate Researcher 2020 - 2024
Florida International University, Miami, FL

- Engineered and implemented innovative, lightweight communication protocols for robot-to-robot interactions, enhancing bandwidth efficiency by 30% and reducing communication overhead in multi-robot systems.
- Addressed and resolved complex privacy issues in multi-robot task allocation and multi-vehicle routing, developing solutions that safeguard sensitive data during autonomous operations.
- Developed a sophisticated motion planning algorithm to identify maximally informative trajectories for decentralized mobile robots, significantly improving data acquisition efficiency and operational autonomy.
- Conducted thorough research on joint computation and task execution by multiple robots without compromising private information, contributing to new standards in robotic collaboration and data security.
- Played a key role in pioneering research initiatives focused on advancing the state of privacy preservation in robotics, leading to potential applications in secure autonomous transport and surveillance systems.
- Mentored and supervised a team of four undergraduate students and two educators, guiding them through complex robotics projects, culminating in two students receiving academic awards for research excellence.

Teaching Assistant and Student Assistant 2020 - 2023
Florida International University, Miami, FL

- Assisted in teaching **CAP 4630 Artificial Intelligence** and **CDA 4625 Introduction to Mobile Robotics**.
- Designed and graded assignments covering topics such as intelligent agents, adversarial search, motion planning, and reinforcement learning.
- Provided individual and group support to students, helping them with programming assignments and course material.
- Facilitated classroom discussions and actively engaged students during lectures and practical sessions.
- Contributed to inclusive learning by addressing diverse student needs, encouraging participation, and fostering a collaborative classroom environment.

AI4ALL Instructor 2023
Miami, FL

- Mentored students in the **Discover AI** program, guiding them through assignments and monitoring their progress.
- Facilitated final student presentations, fostering a sense of accomplishment and confidence in presenting technical material.
- Encouraged discussions on ethical AI, privacy, and diversity in technology, promoting awareness and inclusion.

Automation Engineer 2013 - 2018
Schneider Electric, Saudi Arabia

- Enhanced the security of SCADA systems by implementing robust cybersecurity measures, successfully mitigating potential threats, and reducing cyber-attack vulnerability by over 40%.
- Spearheaded patch management and vulnerability assessments for SCADA systems, reducing system downtime and improving security breach response time by 30%.

Selected Research Contributions

- Designed a motion planning algorithm enabling decentralized robots to find trajectories with maximum information gain.
- Developed a cutting-edge robotic control system, enhancing localization capabilities of surface and aerial robots.
- Published research on privacy preservation in robotics and path planning in reputable conferences and journals.
- Led a team to develop a novel algorithm for privacy-preserving robotic communication, improving data security by 35%.
- Mentored students and educators through robotics projects, resulting in awards for academic excellence.

Research Funding Activities

- Contributed to NSF, USDA-NIFA, and Department of Defense proposal development.
- Participated in multidisciplinary research initiatives involving AI, robotics, autonomous systems, and workforce development.

Selected Publications

- **Alsayegh, M.**, Xu, Y., *A Secure Multi-Party Computation Framework for Decentralized Row Allocation in Cooperative Strawberry Harvesting Systems*, IEEE International Conference on Automation Science and Engineering (CASE), 2025.
- Newaz, A. A. R., **Alsayegh, M.**, Alam, T., Bobadilla, L., *Decentralized Multi-Robot Information Gathering from Unknown Spatial Fields*, IEEE Robotics and Automation Letters (RA-L), vol. 8, no. 5, pp. 3070–3077, 2023.
- **Alsayegh, M.**, Fuentes, J., Bobadilla, L., Shell, D. A., *Oblivious Markov Decision Processes: Planning and Policy Execution*, IEEE Conference on Decision and Control (CDC), 2023.
- **Alsayegh, M.**, Vanegas, P., Newaz, A. A. R., Bobadilla, L., Shell, D. A., *Privacy-Preserving Multi-Robot Task Allocation via Secure Multi-Party Computation*, European Control Conference (ECC), 2022.
- **Alsayegh, M.**, Dutta, A., Vanegas, P., Bobadilla, L., *Lightweight Multi-Robot Communication Protocols for Information Synchronization*, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020.
- **Alsayegh, M.**, Xu, Y., *Decentralized Row Allocation for Harvesting Robots with Uncertain Harvesting Speeds in an Actor–Critic Structure*, Annual Simulation Symposium (ANNSIM), 2026. *Accepted*.
- Soykan, B., **Alsayegh, M.**, Mondesire, S., Rabadi, G., *Differentiable Agent-Based Modeling for Multi-Echelon Supply Chains: Gradient-Based Optimization of Heterogeneous Policies at Scale*, Annual Simulation Symposium (ANNSIM), 2026. *Accepted*.
- Shefin, R., Gupta, D., **Alsayegh, M.**, Alqahtani, S., *When One Agent Reshapes Another: Cross-Hessian Interpretation of Local Coupling in Multi-Agent Reinforcement Learning*, The 4th World Conference on Explainable Artificial Intelligence (XAI 2026), Late-Breaking Work Proceedings (CEUR-WS), 2026. *Accepted*.

Service and Peer Review

- Reviewed submissions at the 2022 European Control Conference (ECC).
- Reviewed submissions at the 2023-2026 IEEE International Conference on Robotics and Automation (ICRA).